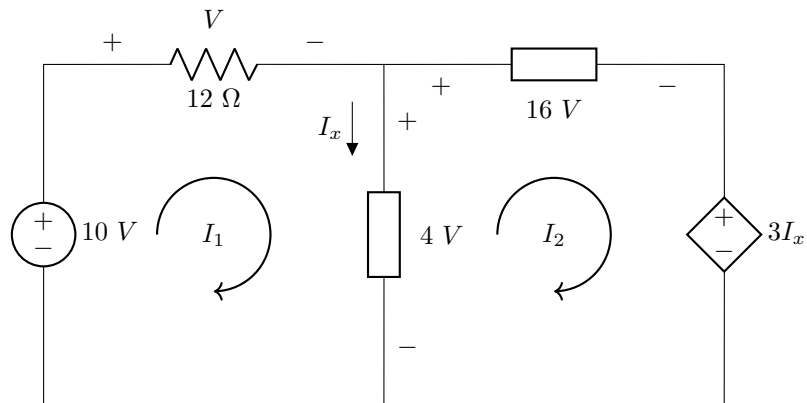


Tutorial 2

Question 2.15

Find V and I_x



Memo:
KVL Eq. 2.19

$$\sum_{m=1}^M V_M = 0$$

Loop I_1 :

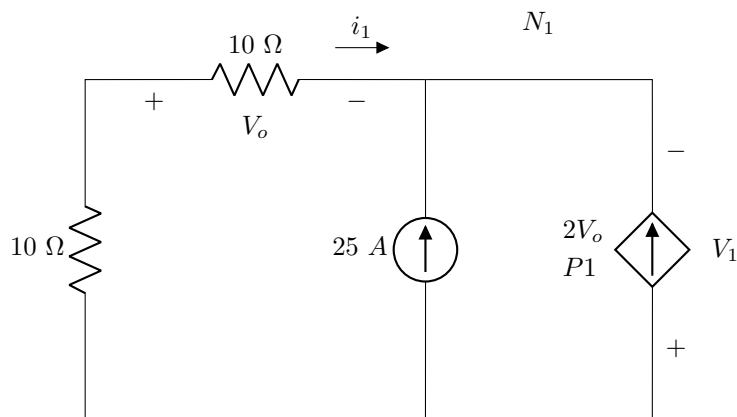
$$\begin{aligned} -10 + V + 4 &= 0 \\ V &= 6V \end{aligned}$$

Loop I_2 :

$$\begin{aligned} -4 + 16 + 3I_x &= 0 \\ 3I_x &= -12 \\ I_x &= -4A \end{aligned}$$

Question 2.22

Find V_o and power over the $2V_o$ source



Memo:

Apply KCL Eq. 2.13

$$\sum_{n=1}^N I_N = 0$$

Part 1:

$$V_o = i_1 R_{10\Omega} \quad (1)$$

From Eq. 2.13

$$\begin{aligned} i_1 + 25 + 2V_o &= 0 \\ i_1 &= -25 - 2V_o \end{aligned} \quad (2)$$

Substitute equation (2) into equation (1)

$$\begin{aligned} V_o &= (-25 - 2V_o) \cdot 10 \\ V_o &= -250 - 20V_o \\ 21V_o &= -250 \\ V_o &= -11.904V \end{aligned}$$

Part 2: From Eq. 1.7

$$\begin{aligned} P &= VI \\ i &= 2V_o \end{aligned}$$

Let R_{eq} be

$$R_{eq} = 10 + 10 = 20\Omega$$

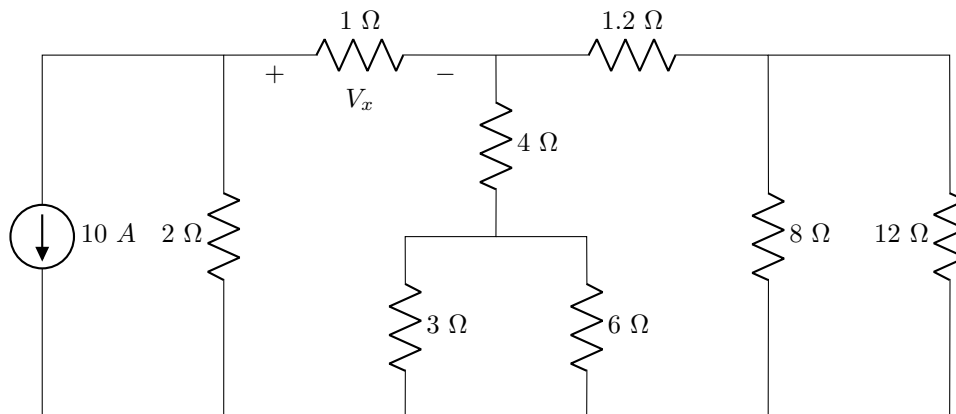
Thus

$$\begin{aligned} V_1 &= i_1 \cdot R_{eq} \\ V_1 &= (-25 - 2(-11.904)) \cdot 20 \\ V_1 &= -23.84V \end{aligned}$$

$$\begin{aligned} P_1 &= V_1 i \\ &= -23.84 \cdot 2(-11.904) \\ &= 566W \end{aligned}$$

Question 2.23

In the circuit below, determine V_x and the power absorbed by the $12\ \Omega$ resistor.



Memo:

From Eq. 2.37

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

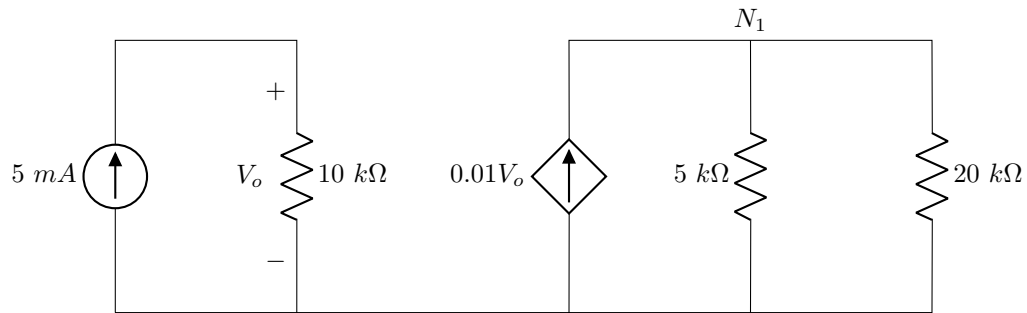
$$12\Omega || 8\Omega = \frac{8 \times 12}{8 + 12} = 4.8\Omega$$

$$6\Omega || 3\Omega = \frac{6 \times 3}{6 + 3} = 2\Omega$$

Redraw:

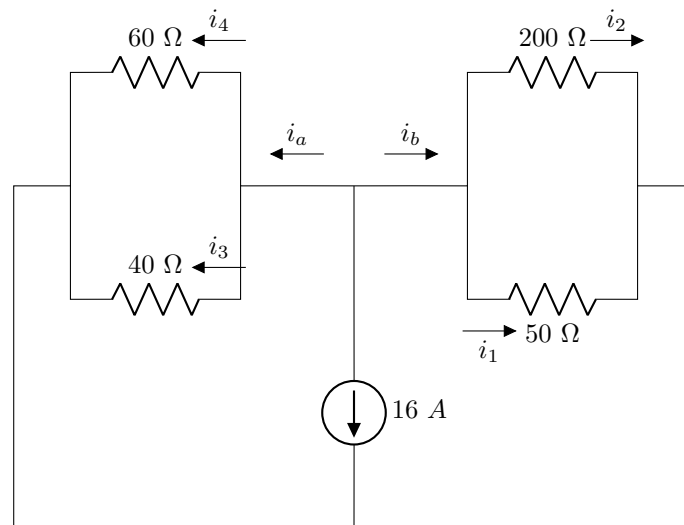
Question 2.25

Find $V_{20k\Omega}$, $i_{20k\Omega}$ and $P_{20k\Omega}$



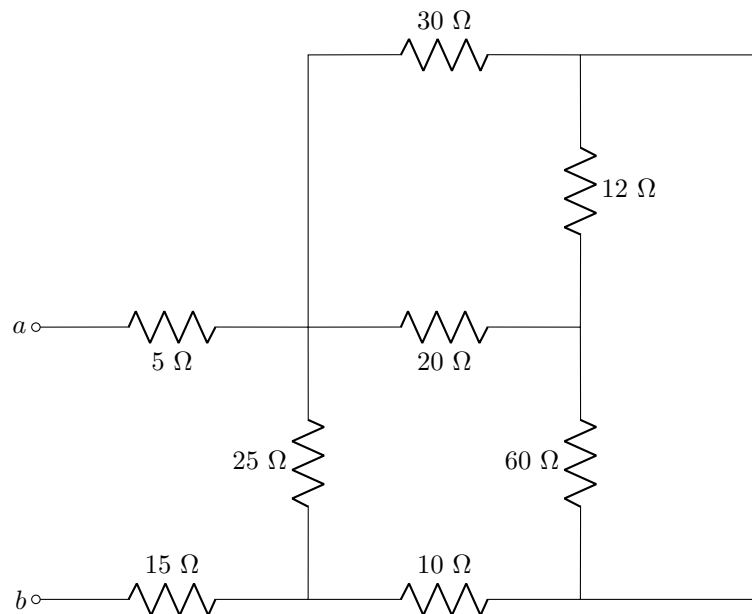
Question 2.32

Find i_1, i_2, i_3 and i_4 .



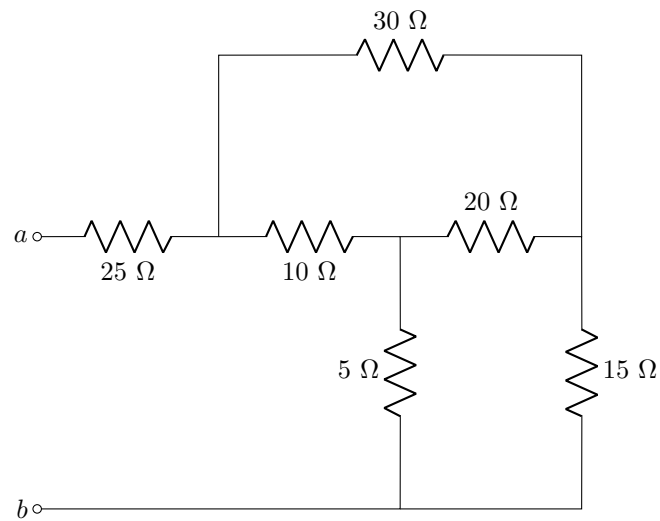
Question 2.45b

Find R_{eq} between terminals $a - b$

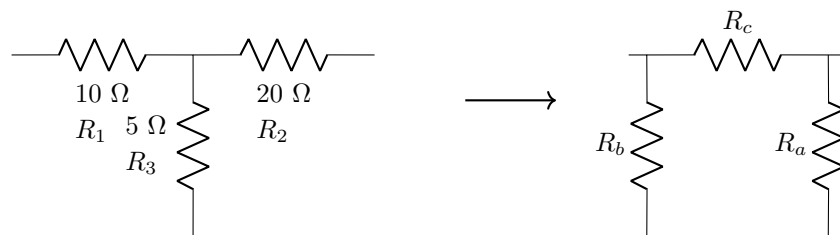


Question 2.51b

Determine the equivalent resistance across $a - b$.



Memo: Perform a y-delta conversion on the middle 3 elements

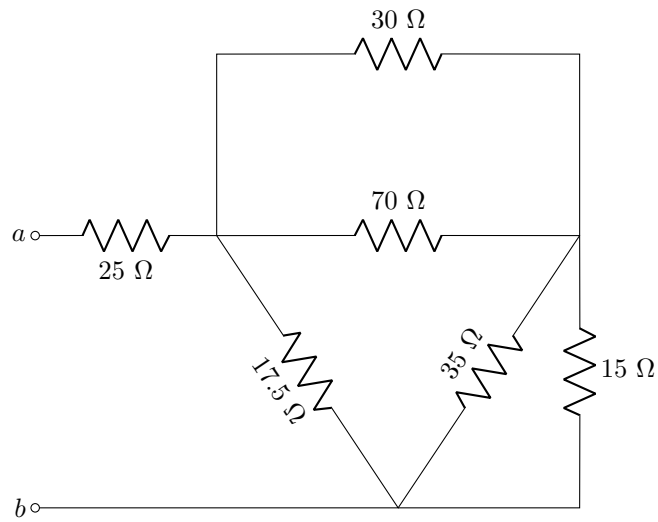


$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = 35 \, \Omega$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = 17.5 \, \Omega$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = 70 \, \Omega$$

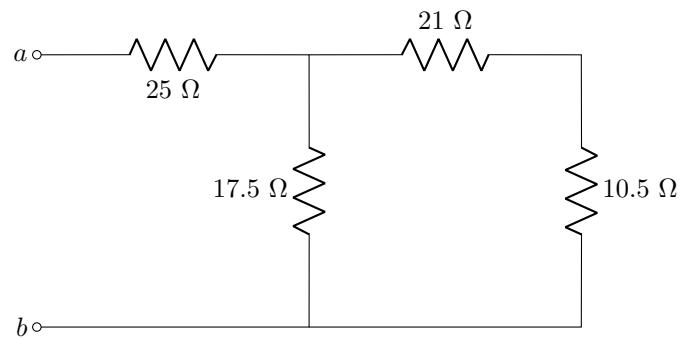
Redraw



$$70 \parallel 30 = 21\ \Omega$$

$$35 \parallel 15 = 10.5\ \Omega$$

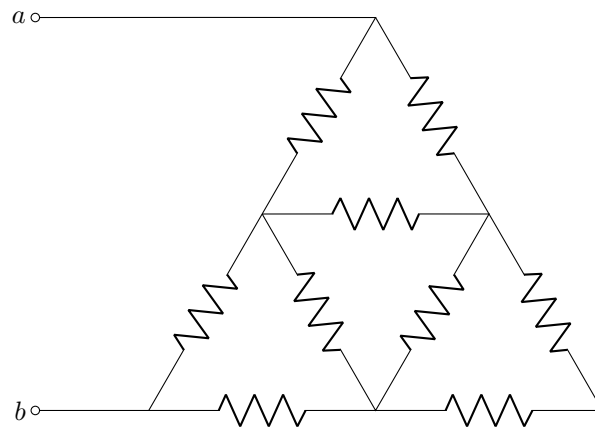
Thus



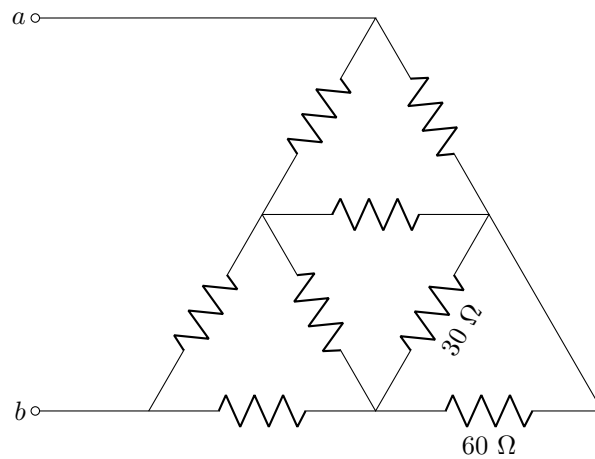
$$R_{ab} = 25 + 17.5 \parallel (21 + 10.5) = 36.25\ \Omega$$

Question 2.53b

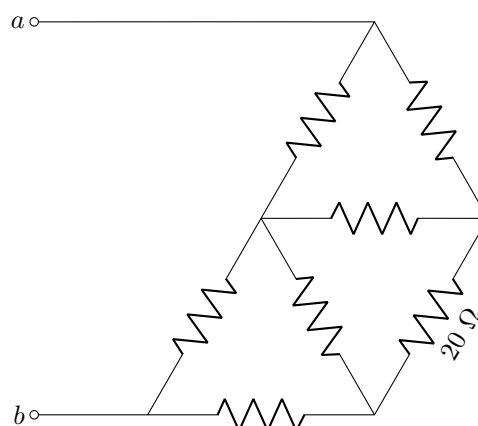
Determine the equivalent resistance across $a - b$ if all the elements are $30\ \Omega$.



Memo: Combine the bottom right 2 series elements to get

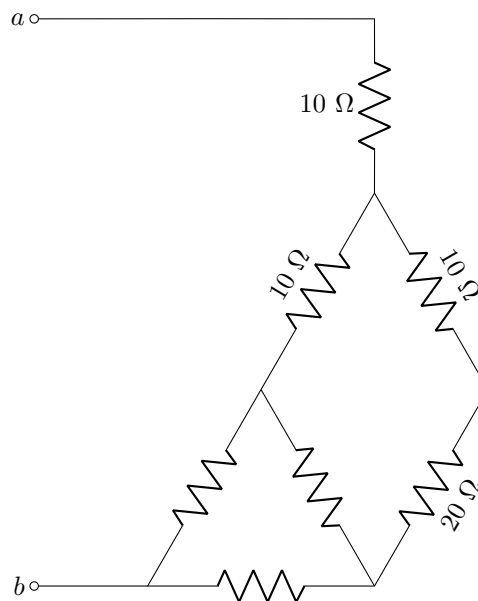


This can be combined with the $30\ \Omega$ in parallel:

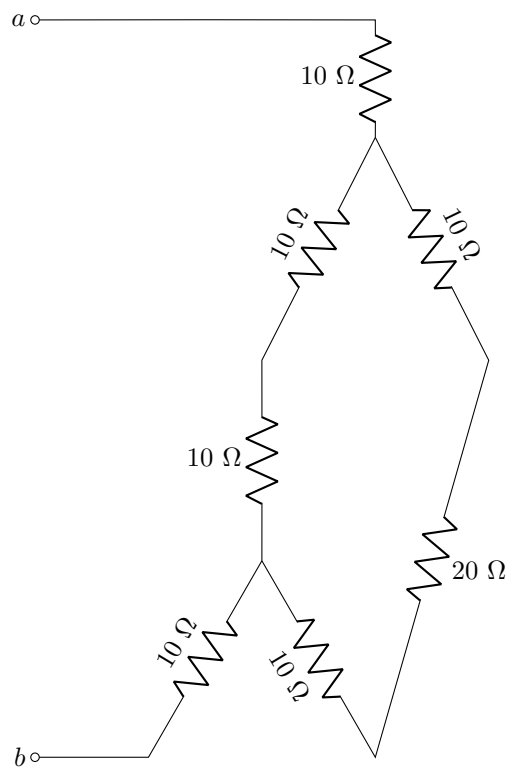


We can now convert the top triangle using y delta

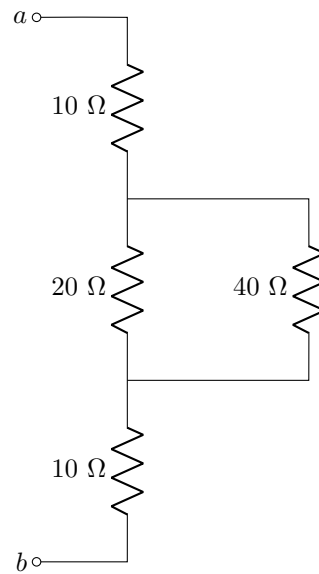
$$R_y = R_{\Delta}/3 = 10 \, \Omega$$



Now we can do the same thing with the bottom left triangle



The 2 10s and the 20 on the right are in series as well as the two tens in the middle. This simplifies to:



Thus:

$$R_{ab} = 10 + 20 || 40 + 10 = 33.33 \Omega$$